



CSA15 · CLOUD COMPUTING AND BIG DATA ANALYTICS

Cloud Storage Technologies

Object, Block, and File Storage Compared

Objective

Understand different cloud storage technologies, compare their architectures, identify their advantages, and explore real-world applications.

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What is Cloud Storage?



Definition

A service that allows users and organizations to store data on remote servers managed by cloud providers instead of local hard drives.

“Cloud storage eliminates the need for physical storage devices by storing data in secure cloud data centers.”

Key Features



Accessible from anywhere



Secure



Highly scalable



Automatic backup



Cost-effective



High availability

Examples

Google Drive

OneDrive

Amazon S3

Dropbox

Types of Cloud Storage

Three major storage technologies are used in cloud computing — each designed for a different workload.



Object Storage

STORES

Files with metadata



Block Storage

STORES

Fixed-size data blocks



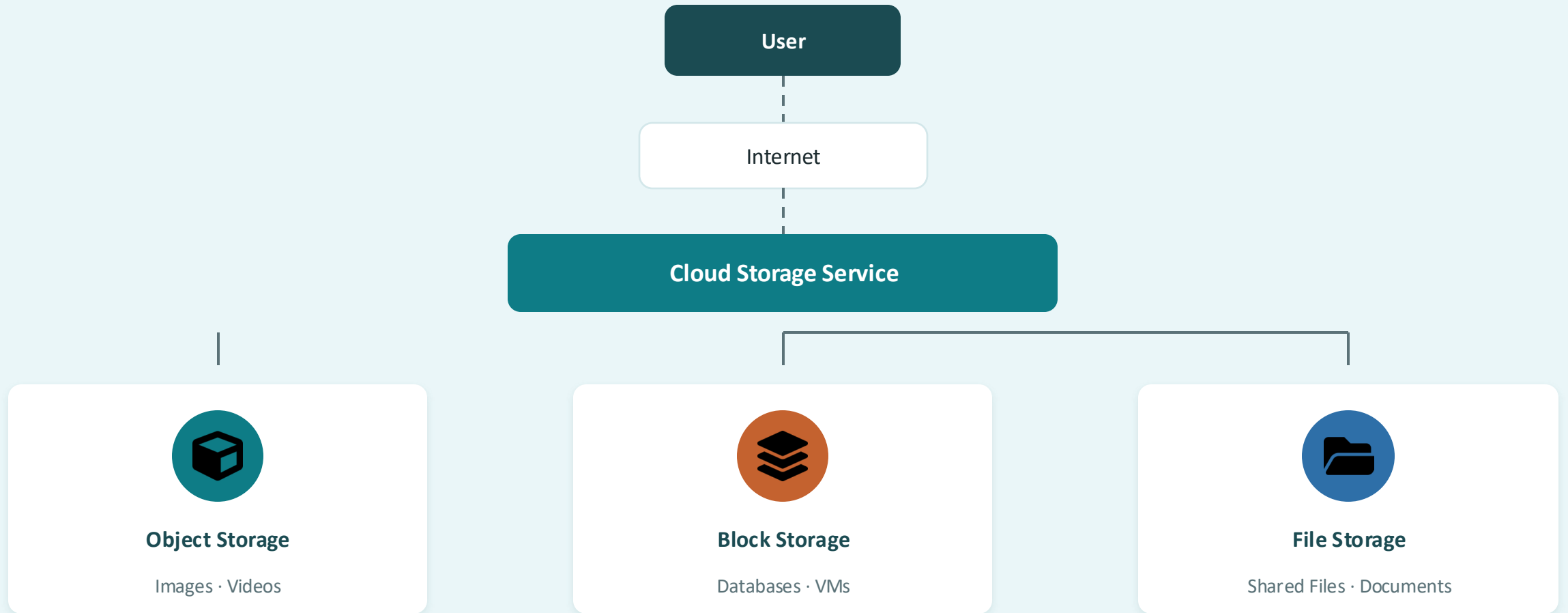
File Storage

STORES

Files inside folders

Architecture Overview

The cloud provider manages storage servers while users access data through applications or APIs.





Object Storage

DEFINITION

Stores complete files as objects — each containing data, metadata, and a unique ID.

Advantages

- ✓ Unlimited scalability
- ✓ Cost-effective
- ✓ Excellent durability
- ✓ Easy backup

Examples

Amazon S3

Azure Blob Storage

Google Cloud Storage

Use Cases

- Photos
- Videos
- Backups
- Data lakes



Block Storage

DEFINITION

Stores data in equal-sized blocks — each block has its own address.

Advantages

- ✓ Very fast
- ✓ Low latency
- ✓ High performance

Examples

Amazon EBS

Azure Managed Disk

Google Persistent Disk

Use Cases

- Virtual Machines
- Databases
- ERP Systems



File Storage

DEFINITION

Stores files inside folders using a familiar hierarchical structure.

Advantages

- ✓ Easy to manage
- ✓ Shared access
- ✓ Familiar file system

Examples

Amazon EFS

Azure Files

Google Filestore

Use Cases

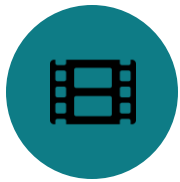
- Team collaboration
- Shared folders
- Enterprise documents

Comparison at a Glance

Feature	Object Storage	Block Storage	File Storage
Structure	Objects	Blocks	Files & Folders
Speed	Medium	Very High	High
Scalability	Excellent	Good	Moderate
Cost	Low	Higher	Medium
Best For	Images, Videos, Backup	Databases, VMs	Shared Documents
Examples	Amazon S3	Amazon EBS	Amazon EFS

Industry Applications

Almost every cloud application combines multiple storage technologies.



Netflix

Uses Object Storage for storing movies and TV content.



Banking

Uses Block Storage for transaction databases.



Companies

Use File Storage for employee documents and collaboration.

Key Takeaways

- ✓ Cloud storage provides scalable and reliable data storage.
- ✓ Object Storage is best for unstructured data.
- ✓ Block Storage is ideal for high-performance applications.
- ✓ File Storage is suitable for shared file access.
- ✓ Choosing the correct storage technology improves performance and reduces costs.

Thank You

Questions?



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